

A NOTE ON ESTIMATING THE RANGE OF A UNIFORM DISTRIBUTION

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Summary: Ghosh and Mukhopadhyay (1975a) and more recently Mukhopadhyay, *et al.* (1983) have considered sequential minimum risk point estimation of θ in a uniform $(0, \theta)$ population. The loss function has been taken as squared error plus cost of sampling, and a new sequential sampling procedure is proposed. However, the corresponding randomly stopped sample maximum is still considered as the final estimator of θ , and this is the major deviation from the usual standard routine. Important theoretical findings have been summarized in the theorem, and it should also be emphasized that the moderate sample size performances of this new and rival procedure do appear to be extremely encouraging.

1. INTRODUCTION

Suppose $X_1, X_2, \dots$ are i.i.d. with the density function $f_\theta(x) = \theta^{-1}I(0 < x < \theta)$, $I(\cdot)$ being the indicator function of $(\cdot)$.

Having observed $X_1, \dots, X_n$, let $T_n = \max\{X_1, \dots, X_n\}$ and $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. We assume that the loss incurred estimating θ by T_n is given by

$$(1.1) \quad L_n = A(\theta - T_n)^2 + cn$$

where $A(>0)$ is the known weight and $c(>0)$ is the known cost per unit sample. The risk associated with (1.1) is given by

$$(1.2) \quad R_n(c) = E(L_n) \\ \approx 2A\theta^2 n^{-2} + cn$$

where we write $a(y) \approx b(y)$ when $a(y)/b(y) \rightarrow 1$ as $y \rightarrow \infty$. Treating n as a continuous variable, we find

$$(1.3) \quad n^* = (4A\theta^2/c)^{1/3}$$

as the expression for n which minimizes the risk in (1.2), the corresponding minimum risk being $R_{n^*}(c)$.

Naturally, n^* is unknown. So, the possibilities of utilizing sequential techniques have been explored in the literature in the past. Ghosh and Mukhopadhyay (1975a) treated the same problem except that they took the absolute error plus cost as the loss function instead. Mukhopadhyay, *et al.* (1983) proposed both sequential and two-stage sampling procedures for the power family distributions of which the uniform distribution is only a special case. Several other relevant citations can be found in Mukhopadhyay, *et al.* (1983). The sequential estimation

procedure proposed in that paper goes like this: Let

$$(1.4) \quad N^* = \inf\{n \geq m: cn^3 \geq 4AT_n^2\}$$

for $m(\geq 1)$ being the starting sample size. When sampling stopped, Mukhopadhyay, *et al.* (1983) estimated θ by T_{N^*} . This estimation procedure was shown to be asymptotically (as $c \rightarrow 0$) risk efficient. However, neither any second-order asymptotic expansion of $E(N^*)$ nor the asymptotic distribution of a suitably normalized version of N^* could be derived in that paper. The reader should also refer to a more recent contribution due to Ghosh and Wackerly (1985).

In section 2, we propose a different sequential estimation procedure which is shown to be asymptotically risk efficient. Moderate sample size performances of this new procedure and the procedure in (1.4) are noticeably about the same. The reader just needs to compare the table given here with table II in Mukhopadhyay, *et al.* (1983). Also, a suitably normalized version of this new stopping variable is shown to be asymptotically normal, and we have given the second-order expansion of the associated average sample size.

2. A NEW SEQUENTIAL PROCEDURE

Recall that $cn^{*3} = 4A\theta^2$. The idea is to use a multiple of $2\bar{X}_n$ to estimate θ while proposing our new stopping variable. More specifically, note that $E\{4(1 + \frac{1}{3n})^{-1} \bar{X}_n^2\} = \theta^2$ and $4(1 + \frac{1}{3n})^{-1} \bar{X}_n^2$ is also consistent for θ^2 . We define

$$(2.1) \quad N = \inf\{n \geq m(\geq 1) : cn^3 \geq 16A(1 + \frac{1}{3n})^{-1} \bar{X}_n^2\}$$

where m is the starting sample size. We can easily show that $P(N < \infty) = 1$ for all $\theta \in (0, \infty)$. When sampling stops using the rule (2.1), we estimate θ by $T_N = \max\{X_1, \dots, X_N\}$. The stopping rule in (2.1) can be equivalently written as

$$(2.2) \quad N = \inf\{n \geq m(\geq 1) : \sum_{i=1}^n X_i \leq d n^{5/2} (1 + \frac{1}{3n})^{1/2}\}$$

where $d = \{c/(16A)\}^{1/2}$. One can easily see that (2.2) has exactly the same form as in Woodroffe's (1977) equation (1.1) with his $\alpha = 5/2$, $L_0 = 1/6$, $c = d$, $\beta = 2/3$, $\mu = \frac{1}{2}\theta$, $\tau^2 = \theta^2/12$, and $\lambda = n^*$. Woodroffe's equation (2.5) holds with $a = 1$. Define

$$\nu = \frac{\beta}{2\mu} \{(\alpha-1)^2 \mu^2 + \tau^2\} - \sum_{n=1}^{\infty} n^{-1} E\left\{\left(\sum_{i=1}^n X_i - n\alpha\mu\right)^+\right\}$$

where $x^+ = \max(x, 0)$. Then, we have

$$\nu\theta^{-1} = \frac{31}{72} - \sum_{n=1}^{\infty} n^{-1} E\left\{\left(S_n - \frac{5n}{4}\right)^+\right\}$$

where $S_n = \sum_{i=1}^n Y_i$, Y_i 's being i.i.d. uniform on $(0, 1)$. It is obvious that $S_n \leq n$ w.p.1 and hence $(S_n - \frac{5n}{4})^+ = 0$ w.p.1. Thus, $\nu\theta^{-1} = 31/72$. Now, the risk associated with our sequential procedure (2.1) is given by

$$\begin{aligned}
 (2.3) \quad R(c) &= E(L_N) \\
 &= AE\{(\theta - T_N)^2\} + cE(N).
 \end{aligned}$$

Next, we define the "risk-efficiency" of our procedure as $\eta(c) = R(c)/R_{n^*}(c)$ and the corresponding "regret" being $\omega(c) = R(c) - R_{n^*}(c)$. The main asymptotic findings are now summarized in the following theorem.

Theorem: For the sequential procedure (2.1) we have the following as $c \rightarrow 0$:

- (i) $n^{*-1/2}(N - n^*) \xrightarrow{L} N(0, \frac{4}{27})$,
- (ii) $E(N) = n^* + \frac{5}{18} + o(1)$ for $m \geq 1$,
- (iii) $E\{(N - E(N))^2\} = \frac{4}{27} n^* + o(n^*)$ for $m \geq 1$,
- (iv) $n^{*2}E\{(\theta - T_N)^2\} = 2\theta^2 + o(1)$ for $m \geq 2$,
- (v) $\eta(c) \rightarrow 1$ for $m \geq 2$.

Proof: Part (i) follows from the theorem of Ghosh and Mukhopadhyay (1975b). See also theorem 3 in Ghosh and Mukhopadhyay (1979).

From Woodroffe's (1977) theorem 2.4 and the representation (2.2), we can verify the following for $m \geq 1$:

$$\begin{aligned}
 E(N) &= n^* + \frac{4}{3} \nu \theta^{-1} - \frac{8}{27} + o(1) \\
 &= n^* + \frac{5}{18} + o(1).
 \end{aligned}$$

This is part (ii).

Part (iii) is immediate if we combine part (i) with the

theorem 2.3 from Woodroffe (1977).

Let us now prove part (iv). For all $r \geq 2$ and $0 < \phi$, $r < 1$ we obtain

$$(2.4) \quad P\{N \leq \phi n^*\} = O(c^{1/2m}) + O(c^{r\gamma/6})$$

from Woodroffe's (1977) lemma 2.3. From Anscombe's (1952) result it follows that $n^*(\theta - T_N)/\theta \xrightarrow{L} Y$ as $c \rightarrow 0$ where Y is a standard exponential random variable. So, to prove part (iv), it is sufficient to prove that $Z_N = n^{*2}(\theta - T_N)^2$ is uniformly integrable. This can be verified by showing that $\sup_{0 < c < \infty} E(Z_N^{1+\epsilon})$ is bounded for some $\epsilon > 0$. See, for example, part (ii) in lemma A on page 13 of Serfling (1980). Now, for $\epsilon > 0$, we have

$$\begin{aligned} E\{Z_N^{1+\epsilon}\} &= n^{*2(1+\epsilon)} E\{(\theta - T_N)^{2+2\epsilon}\} \\ &\leq (n^*\theta)^{2+2\epsilon} [P(N \leq \phi n^*) + E\{(1 - \theta^{-1}T_N)^{2+2\epsilon}\}] \\ &= I + II, \end{aligned}$$

say, where $\langle x \rangle$ stands for the largest integer $\leq x$. The term II is $O(1)$ and we can show this by exploiting properties of gamma functions as it was done in equation (2.14) of Mukhopadhyay, *et al.* (1983). The term I , in view of (2.4), can be shown to be $O(1)$ if we choose $\epsilon = \frac{1}{4}$ and $m \geq 2$.

To prove part (v), first notice that $R_{n^*}(c) \approx 3cn^*/2$. Then, we need to combine parts (ii) and (iv) with our (2.3), and the rest would follow as in Mukhopadhyay, *et al.* (1983).

This completes the proof.

Remark 1: Equation (2.4) compares favourably with part (iv) in lemma 1 of Mukhopadhyay, *et al.* (1983). Part (ii) in our theorem is much stronger than the conclusion in part (iii) of the same lemma in Mukhopadhyay, *et al.* (1983). Part (v) shows that the sequential procedure (2.1) is indeed "asymptotically risk-efficient" which was exactly the same finding in theorem 2 of Mukhopadhyay, *et al.* (1983) for their stopping rule which has been restated here as (1.4).

Remark 2: Regarding the possibility of having an asymptotic distribution of a normalized version of Mukhopadhyay, *et al.*'s (1983) stopping time, only some partial results were obtained in theorem 3 in that paper. Part (i) in our theorem is definitely more encouraging and moreover we are able to give an expansion for the expected value and variance of our stopping time in parts (ii) and (iii).

In spite of the fact that the purely sequential procedure (2.1) has most of the "good" properties enjoyed by the sequential procedure of Mukhopadhyay, *et al.* (1983) and more (look at properties (i) - (iii) in our theorem), there will undoubtedly be criticisms simply because we have basically used $4\bar{x}_n^2$ to estimate θ^2 in (2.1) as opposed to $4T_n^2$. Probably, one would say that Mukhopadhyay, *et al.*'s (1983) sequential procedure (2.7) is "better" than the sequential procedure (2.1) as proposed here in some sense. But, this whole issue is still up in the air. At this stage, it seems to be extremely interesting to find out that the stopping rule (2.1) does so well compared to a rival procedure that is based on $\{T_n\}$, the sequence of sufficient statistics. Since the nonlinear renewal theory is not yet developed enough to

give us the second-order approximations of $\omega(c)$ for the two rival procedures in question, we settle for the moderate sample size performances for purposes of some *ad hoc* comparisons.

3. MODERATE SAMPLE-SIZE PERFORMANCE

In this section, we study briefly the performance of the new sequential procedure (2.1) when $n^* = 10(5)30, 50(50)200$. We fix $\theta = 1$ and generate pseudorandom uniform deviates using a FORTRAN program on an IBM 370/168 system with WATV compiler. We take $A = 1$ and thus $c = 4/n^{*3}$, and we also consider $m = 3, 5, 10$. That is, we start with m observations and proceed sequentially until the rule given in (2.1) indicates that sampling terminates. In each instance, we have repeated this procedure a total of 500 times. Each time we ceased sampling, we determined the observed value of N , say $n(i)$, the observed value of T_N , $T_{n(i)}$, and $L_{n(i)} = (1 - T'_{n(i)})^2 + cn(i)$. Throughout our table, we use the following notations:

$$\bar{n} = k^{-1} \sum_{i=1}^k n(i), \quad s^2(\bar{n}) = \{k(k-1)\}^{-1} \sum_{i=1}^k (n(i) - \bar{n})^2,$$

$$\bar{T} = k^{-1} \sum_{i=1}^k T_{n(i)}, \quad s^2(\bar{T}) = \{k(k-1)\}^{-1} \sum_{i=1}^k (T_{n(i)} - \bar{T})^2,$$

$$\bar{L} = k^{-1} \sum_{i=1}^k L_{n(i)}, \quad U_{n(i)} = \text{the observed value of } 2\bar{X}_N$$

$$\text{when } N = n(i), \quad \bar{U} = k^{-1} \sum_{i=1}^k U_{n(i)},$$

$$s^2(\bar{U}) = \{k(k-1)\}^{-1} \sum_{i=1}^k (U_{n(i)} - \bar{U})^2 \quad \text{for } i = 1, \dots, k$$

and $k = 500$.

We estimate $\eta(c)$ and $\omega(c)$ by $\hat{\eta}(c) = \bar{L}/R_{n^*}(c)$
and $\hat{\omega}(c) = \bar{L} - R_{n^*}(c)$.

Remark 3: We notice almost no variation in the performances for different values of m . Thus, $m = 3$ seems to be the right choice from the conservative point of view.

Remark 4: $\bar{U}$ seems to be nearer to $\theta = 1$ compared to the more conventional one, namely, $\bar{T}$. Thus, surprisingly enough, $2\bar{X}_N$ does perform quite well compared with T_N as rival estimators for θ .

Remark 5: Comparing our $\hat{\eta}$ and $\hat{\omega}$ with those in table II of Mukhopadhyay, *et al.* (1983), one can see that the new sequential procedure (2.1) performs somewhat better for most of the entries. At the same time, we also have nice second-order expansions for $E(N)$ and $V(N)$ as presented in our theorem. Thus, one has more insight into our new stopping rule (2.1) and this in turn gives the experimenter somewhat better handle of the sampling procedure. The procedure (2.1) is very appealing with sound theoretical properties and it also performs extremely well for moderate values of n^* . We recommend its use in practice very strongly.

Remark 6: Comparing $s(\bar{n})$ and $s(\bar{T})$ with $(500)^{1/2}SD(n)$ and $(500)^{1/2}SD(T)$ in table II of Mukhopadhyay, *et al.* (1983), one can see that our $\bar{n}$ and $\bar{T}$ are indeed more variable.

TABLE
Performance of the Sequential Procedure (2.1)

n^*	m	$\bar{n}$	$s(\bar{n})$	$\bar{T}$	$s(\bar{T})$	$\bar{U}$	$s(\bar{U})$	$\hat{\eta}$	$\hat{\omega} \times 10^5$
10	3	10,23	0,062	0,897	0,005	0,970	0,009	1,073	440
	5	10,23	0,062	0,897	0,005	0,970	0,009	1,070	419
	10	10,65	0,035	0,908	0,004	0,984	0,008	0,984	-99
15	3	15,13	0,078	0,927	0,004	0,974	0,007	1,118	315
	5	15,16	0,074	0,932	0,003	0,976	0,007	1,032	86
	10	15,17	0,074	0,932	0,003	0,976	0,007	1,034	91
20	3	20,24	0,083	0,949	0,002	0,986	0,006	1,011	17
	5	20,25	0,083	0,949	0,002	0,986	0,006	1,010	16
	10	20,21	0,083	0,948	0,002	0,986	0,006	1,049	74
25	3	25,20	0,092	0,959	0,002	0,988	0,005	1,001	1
	5	25,22	0,092	0,959	0,002	0,988	0,005	1,006	6
	10	25,23	0,091	0,960	0,002	0,988	0,005	0,999	-1
30	3	30,38	0,094	0,969	0,001	0,998	0,005	0,961	-26
	5	30,39	0,094	0,969	0,001	0,998	0,005	0,960	-27
	10	30,38	0,098	0,968	0,002	0,998	0,005	1,004	3
50	3	50,10	0,120	0,979	0,001	0,990	0,004	1,016	4
	5	50,10	0,120	0,979	0,001	0,990	0,004	1,016	4
	10	50,10	0,120	0,979	0,001	0,990	0,004	1,016	4
100	3	100,56	0,173	0,990	0,000	1,002	0,003	1,017	1
	5	100,56	0,173	0,989	0,000	1,002	0,003	1,017	1
	10	100,56	0,173	0,990	0,000	1,002	0,003	1,017	1
150	3	150,36	0,201	0,993	0,000	0,999	0,002	1,017	1
	5	150,36	0,201	0,993	0,000	0,999	0,002	1,017	1
	10	150,36	0,201	0,993	0,000	0,999	0,002	1,017	1
200	3	199,94	0,238	0,995	0,000	0,996	0,002	1,011	0
	5	199,94	0,238	0,995	0,000	0,996	0,002	1,011	0
	10	199,94	0,238	0,995	0,000	0,996	0,002	1,011	0

However, for some unknown reasons, this is not reflected when we compare $\hat{\eta}$ and $\hat{\omega}$ for the two rival procedures. This brings us back to the same types of comments we had made in the last paragraph of section 2. It would indeed be very challenging and interesting to compare the two rival purely sequential procedures through the second-order expansions of $\omega(c)$. We pose this as an open problem; however, we take this opportunity to conjecture that Mukhopadhyay, *et al.*'s (1983) procedure will come out ahead in a big way. This remains for the future to unfold, and at present friendly rivalries continue.

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